

divvier3 vs divvier3bSnap...
Spreadsheet

Can you determine the statistical significance (or otherwise) of the runtimes measured for 8 input classes for my programs 'divvier3' and 'divvier3b Snapshot4(3)' based on the attached spreadsheet? The latter can be referred to as 'divvier3b(s5). (I have already calculated means and SDs, but the raw data is represented by the sets of 10 measurements for each).

Stopped thinking >

Yes. Since you have 10 independent runtime measurements for each program on each input class, a comparison of the raw measurements is more informative than comparing only the means and SDs.

A suitable test is **Welch's two-sample t-test**, which does not assume equal variances. (With only 10 measurements per group, this is generally preferable to the classical Student t-test.)

Here are the results I obtained from your spreadsheet:

Input	divvier3 mean	divvier3b(s5) mean	Difference	Welch t	p-value	Significance ($\alpha = 0.05$)?
1	6.419	0.000	divvier3b faster	29.24	3.12×10^{-10}	Yes
2	6.619	0.285	divvier3b faster	20.11	1.71×10^{-11}	Yes
3	6.492	0.403	divvier3b faster	25.54	1.44×10^{-15}	Yes
4	7.906	9.040	divvier3 faster	-3.51	0.00253	Yes
5	6.698	0.833	divvier3b faster	20.12	1.07×10^{-11}	Yes
6	81.500	8.250	divvier3b faster	50.37	3.93×10^{-13}	Yes

Input	divvier3 mean	divvier3b(s5) mean	Difference	Welch t	p-value	Significa ($\alpha = 0.05$)?
7	478.058	30.403	divvier3b faster	58.58	4.26×10^{-13}	Yes
8	8.013	9.172	divvier3 faster	-2.28	0.0350	Yes

Interpretation

Every one of the eight comparisons reaches statistical significance at the conventional 5% level.

The practical picture is even clearer:

- **Inputs 1, 2, 3, 5, 6 and 7:** the evidence that **divvier3b(s5)** is faster is overwhelming (p-values between about 10^{-10} and 10^{-15}). These are not borderline results.
- **Input 4:** **divvier3** is significantly faster (about 1.13 runtime units, $p \approx 0.0025$).
- **Input 8:** **divvier3** is also faster, although this is the weakest result ($p \approx 0.035$). It is statistically significant at the 5% level, but would not remain significant under a very stringent threshold.

Multiple comparisons

Because you've performed eight separate tests, it's worth considering a multiple-comparison correction.

Using the **Bonferroni correction**, the significance threshold becomes

$$\alpha = \frac{0.05}{8} = 0.00625.$$

Under this stricter criterion:

- Inputs **1–7 except 8** remain significant.
- **Input 8 ($p = 0.035$)** no longer reaches significance.

So the evidence for Input 8 is relatively weak compared with the others.

Overall conclusion

Your data provide very strong evidence that the two implementations have different runtime characteristics. The snapshot version **divvier3b(s5)** is dramatically faster on six of the eight input classes, especially on the larger or more demanding cases (Inputs 6 and 7). It appears to incur some overhead that makes it slightly slower on Inputs 4 and 8, with the evidence for Input 4 being fairly strong and for Input 8 being comparatively modest.

If you're writing this up for a report or paper, I'd recommend reporting the **Welch t-test** results together with **effect sizes** (for example, Cohen's d or Hedges' g) and **95% confidence intervals** for the mean differences. Those would quantify not just whether the differences are statistically significant, but how large they are in practical terms.

